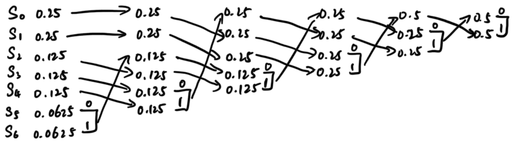
1. $\forall q_1, q_2, \dots, q_N \in [0,1]$, $\sum_{i=1}^{N} q_i = 1$

$\sum_{i=1}^{N} p_i \log\left(\frac{q_i}{p_i}\right) \le \log e \sum_{i=1}^{N} p_i\left(\frac{q_i}{p_i} - 1\right) = \log e \sum_{i=1}^{N}(q_i - p_i) = 0$

$\Rightarrow H(X) = \sum_{i=1}^{N} p_i \log\left(\frac{1}{p_i}\right) \le \sum_{i=1}^{N} p_i \log\left(\frac{1}{q_i}\right)$ (iff $p_i = q_i$, $i = 1,2,3,\dots N$)

取 $q_i = \frac{1}{N}$，可得 $H(X) \le \log N$ (iff $p_i = \frac{1}{N}$, $i = 1,2,\dots N$)

2. (a)



S_0 10
S_1 11
S_2 001
S_3 010
S_4 011
S_5 0000
S_6 0001

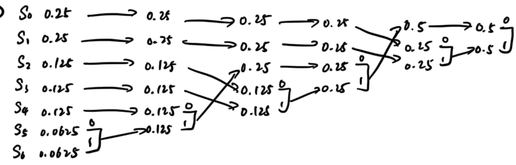
$\bar{L} = 0.25\times 2\times 2 + 0.125\times 3\times 3 + 0.0625\times 4\times 2$

$= 2.625$

$var(L) = 0.25\times 1^2\times 2 + 0.125\times 2^2\times 3 + 0.625\times 4^2\times 2 - 2.625^2$

$= 7.375 - 2.625^2 = 0.484375$

(b)



S_0 10
S_1 11
S_2 010
S_3 011
S_4 000
S_5 0010
S_6 0011

$\bar{L} = 2.625$

$var(L) = 0.484375$

(c) $H(S) = 0.25\times\log\frac{1}{0.25}\times 2 + 0.125\times\log\frac{1}{0.125}\times 3 + 0.0625\times\log\frac{1}{0.0625}\times 2$

$= 2.625 = \bar{L}$

$\Rightarrow \eta = \frac{H(S)}{\bar{L}}$

因为 $l_k = \log\frac{1}{p_k}$, $k = 0,1,\dots 6$，所以码字最佳

(d) $var(L)\big|_a = var(L)\big|_b$，证明两种 Huffman Code 是等效的

3. $H(X_1 X_2 \cdots X_k) = -\sum_{n_1}\sum_{n_2}\cdots\sum_{n_k} P_{n_1 n_2 \cdots n_k} \log P_{n_1 n_2 \cdots n_k}$

$= -\sum_{n_1}\sum_{n_2}\cdots\sum_{n_k} P_{n_k|n_1\cdots n_{k-1}} P_{n_1\cdots n_{k-1}} \left(\log P_{n_k|n_1\cdots n_{k-1}} + \log P_{n_1\cdots n_{k-1}}\right)$

$= -\sum_{n_k} P_{n_k|n_1\cdots n_{k-1}} \sum_{n_1}\cdots\sum_{n_{k-1}} P_{n_1\cdots n_{k-1}} \log P_{n_1\cdots n_{k-1}} - \sum_{n_1}\sum_{n_2}\cdots\sum_{n_k} P_{n_1 n_2\cdots n_k} \log P_{n_k|n_1\cdots n_{k-1}}$

$= H(X_1 X_2 \cdots X_{k-1}) + H(X_k | X_{k-1}\cdots X_1)$

$\Leftrightarrow H(X_k\cdots X_1) - H(X_{k-1}\cdots X_1) = H(X_k | X_{k-1}\cdots X_1)$

$\Rightarrow \sum_{n=2}^{k}\left[H(X_n\cdots X_1) - H(X_{n-1}\cdots X_1)\right] = \sum_{n=2}^{k} H(X_n | X_{n-1}\cdots X_1)$

$\Leftrightarrow H(X_k\cdots X_1) = H(X_1) + \sum_{n=2}^{k} H(X_n | X_{n-1}\cdots X_1) = \sum_{i=1}^{k} H(X_i | X_{i-1}\cdots X_1)$

4. (a) $H(XY) = H(X) + H(Y|X) = H(X) + H(X^2|X) = H(X) = \log 5$

(b) $H(Y|X) = H(X^2|X) = 0$

(c) $H(X|Y) = H(XY) - H(Y) = H(X) - H(Y) = \log 5 - \left(0.4\times\log\frac{1}{0.4}\times 2 + 0.2\times\log\frac{1}{0.2}\right)$

$= \log 5 - (\log 5 - 0.8) = 0.8$

(d) $I(X;Y) = H(X) - H(X|Y) = H(X) - (H(XY) - H(Y)) = H(Y) = H(XY) - H(X|Y) = \log 5 - 0.8$

5. a) $\forall a \in \mathbb{R}$ 且 $a \neq 0$, $p(ax)\,|a|\,dx = p(x)\,dx$

$h(aX) = -\int_{-\infty}^{+\infty} \log p(ax)\cdot p(ax)\,|a|\,dx$

$= -\int_{-\infty}^{+\infty} \log\left(\frac{p(x)}{|a|}\right) p(x)\,dx$

$= h(X) + \int_{-\infty}^{+\infty} \log|a|\cdot p(x)\,dx$

$= h(X) + \log|a|$

b) $\int_{-\infty}^{+\infty} p(x)\log\frac{q(x)}{p(x)}\,dx \le \log e \int_{-\infty}^{+\infty} p(x)\left(\frac{q(x)}{p(x)} - 1\right)dx = \log e\left(\int_{-\infty}^{+\infty}(q(x) - p(x))\,dx\right) = 0$

$\Leftrightarrow h(X) \le -\int_{-\infty}^{+\infty} p(x)\log q(x)\,dx$ (iff $\frac{q(x)}{p(x)} = 1$, $\forall x \in \mathbb{R}$)

c) 取 $q(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}}$

由 b) 得 $h(X) \le \int_{-\infty}^{+\infty} p(x)\log\left(\sqrt{2\pi}\sigma\, e^{\frac{x^2}{2\sigma^2}}\right)dx$

$= \frac{1}{2}\log(2\pi\sigma^2) + \int_{-\infty}^{+\infty}\frac{x^2}{2\sigma^2}p(x)\,dx$

$= \frac{1}{2}\log(2\pi\sigma^2) + \frac{1}{2\sigma^2}E(X^2)$

$\le \frac{1}{2}\log(2\pi\sigma^2) + \frac{1}{2} = \frac{1}{2}\log(2\pi e\sigma^2)$ (iff $p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}}$)